bigger than the expected energy contribution. Like the comrades in the warehouse entrusted with denominations larger than fifty-dollar bills, these waves with ever-larger frequencies cannot contribute the amount of energy demanded by nineteenth-century physics. And so, just as only a finite number of comrades are able to contribute to the total heat payment—leading to a finite amount of total money—only a finite number of waves are able to contribute to the oven's total energy—again leading to a finite amount of total energy. Be it energy or money the lumpiness of the fundamental units—and the ever increasing size of these lumps as we go to higher frequencies or to larger monetary denominations—changes an infinite answer to one that is finite.3

By eliminating the manifest nonsense of an infinite result, Planck had taken an important step. But what really made people believe that his guess had validity is that the finite answer that his new approach gave for the energy in an oven agreed spectacularly with experimental measurements. Specifically, Planck found that by adjusting one parameter that entered into his new calculations, he could predict accurately the measured energy of an oven for any selected temperature. This one parameter is the proportionality factor between the frequency of a wave and the minimal lump of energy it can have. Planck found that this proportionality factor—now known as Planck's constant and denoted ħ (pronounced "h-bar")—is about a billionth of a billionth of a billionth in everyday units.4 The tiny value of Planck's constant means that the size of the energy lumps are typically very small. This is why, for example, it seems to us that we can cause the energy of a wave on a violin string—and hence the volume of sound it produces—to change continuously. In reality, though, the energy of the wave passes through discrete steps, à la Planck, but the size of the steps is so small that the discrete jumps from one volume to another appear to be smooth. According to Planck's assertion, the size of these jumps in energy grows as the frequency of the waves gets higher and higher (while wavelengths get shorter and shorter). This is the crucial ingredient that resolves the infinite-energy paradox.

As we shall see, Planck's quantum hypothesis does far more than allow us to understand the energy content of an oven. It overturns much about the world that we hold to be self-evident. The smallness of ħ confines most of these radical departures from life-as-usual to the